

AMY

Workforce Financial Operating System for Informal Economies

Version: 2.0 (Revised)

Date: 02nd May 2026

1. PRODUCT OVERVIEW

1.1 Purpose

AMY is a configurable workforce, earnings, and financial management platform designed for **any informal or semi-formal work arrangement** where:

- Work is organized into **assignments** (shifts, days, hours, tasks, deliveries)
- Payment is variable, periodic, or performance-based
- Workers need a **verifiable financial identity** to access credit, insurance, and other services

AMY serves transport operators, construction firms, logistics companies, agricultural estates, community health programs, and any organization managing a distributed or casual workforce across Africa.

It enables:

- Digital management of workforce operations and assignments
 - Real-time earnings tracking at assignment level (shift, day, hour, task)
 - Automated payouts (daily, weekly, monthly) into a digital wallet
 - Seamless statutory compliance and payroll processing
 - Creation of a verifiable financial identity layer
 - Access to credit, insurance, and other financial services
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Core Product Principle (Industry-Agnostic)

text

IDENTITY → ASSIGNMENT → EARNINGS → WALLET → DEDUCTIONS → FINANCIAL PROFILE

This principle applies equally to:

- A matatu driver completing a shift
 - A construction worker clocking hours on a road site
 - A delivery rider completing 20 drops
 - A community health promoter making 15 home visits
 - A booking agent facilitating 10 customer reservations
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1.2 Problem Statement

Millions of informal and semi-formal workers across Africa—in transport, construction, logistics, agriculture, hospitality, and community health—have **no verifiable record of their work or earnings**. They are paid in cash or through untraceable channels. They cannot prove income for loans, insurance, housing, or formal financial services.

The ecosystem suffers from:

- Unstructured, cash-based, and untraceable earnings
- No linkage between work performed and income received
- Lack of transparency for workers, employers, and financiers
- Poor statutory compliance (SHA, NSSF, Housing Levy) where applicable
- No unified system linking **assignments** → **earnings** → **payments** → **compliance** → **financial services**
- Exclusion from credit and insurance due to lack of financial history

For employers (SACCOs, contractors, logistics firms, estates):

- Inefficient manual payroll for casual and shift-based workers
 - Difficulty tracking attendance and hours across multiple sites
 - Limited ability to offer workers financial benefits
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1.3 Solution

AMY provides:

- A **centralized identity & workforce registry** (any worker type)
- A **configurable assignment & earnings engine** (the core innovation — works across industries)
- **Automated wallet-based payouts** via JamboPay
- **Payroll & statutory compliance** via Perpay (configurable per worker type)
- A **financial data layer** powering lending, insurance, and savings
- **Multi-channel access:**
 - Mobile App (smartphones)
 - USSD (feature phones)
 - Web Dashboard (employers/admins)
 - APIs (partners, lenders, insurers)

Key differentiator: AMY is industry-agnostic and tenant-configurable. The same core system powers a matatu driver, a road construction laborer, and a community health promoter — with different assignment types, earning models, and payment frequencies configured at the tenant level.

2. GOALS & OBJECTIVES

2.1 Business Goals

- Digitize workforce operations across **multiple informal sectors**
- Become the **financial operating system** for Africa's informal workers
- Enable **embedded financial services** (loans, insurance, savings)
- Drive **high-volume transaction processing** across industries

2.2 Product Goals

- Enable **real-time earnings visibility** for any assignment type
 - Support **daily, weekly, and monthly automated payouts**
 - Ensure **traceability of earnings to work performed**
 - Achieve **automated statutory compliance** (where applicable)
 - Build **credible financial identities** for informal workers
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2.3 Success Metrics (KPIs)

Category	Metrics
Adoption	# of tenants (SACCOs, contractors, firms), # of active workers
Financial	Transaction volume, payout success rate (>98%)
Earnings	% of earnings traceable to assignments
Compliance	% of eligible workers contributing to SHA/NSSF
Credit	Loan volume, repayment rate (>85%)
Engagement	Daily active workers across all industries

3. USERS & PERSONAS

Persona	Industry Example	Core Actions
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Tenant Admin	SACCO manager, Construction site manager, Logistics dispatcher	Assign workers, fund wallet, run payroll, monitor compliance
Worker	Driver, laborer, rider, promoter, agent	Track earnings, withdraw funds, access loans, view assignments
Lender	Bank, MFI, SACCO	Evaluate creditworthiness, disburse loans, track repayment
Insurance Provider	Insurer	Enroll workers, collect premiums, process claims
System Admin	Platform operator	Monitor system, manage tenants, audit logs

Expanded Worker Personas (By Industry)

Industry	Worker Type	Assignment Model	Payment Frequency
Transport	Driver, Conductor	Per shift + commission	Daily
Construction	General labor, Operator	Hourly or daily fixed	Daily/Weekly
Logistics	Delivery rider	Per delivery	Daily
Agriculture	Harvester, Packer	Per unit (kg, crate)	Weekly
Health	Community Health Promoter	Per visit + monthly stipend	Monthly
Facilitator	Booking agent, Foreman, Supervisor	Commission, salary, hybrid	Weekly/Monthly

4. CORE SYSTEM MODULES

4.1 Identity & Registry (Industry-Agnostic)

- Unique worker ID across multiple tenants (if a worker serves multiple employers)
- KYC verification (IPRS integration where available)
- Role management (configurable per tenant/industry)

- Multi-tenant association (worker can work for SACCO + construction firm)

4.2 Assignment & Operations Engine (Configurable)

Generic assignment model:

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Worker ↔ Work Unit ↔ Location/Project ↔ Duration

Configurable assignment types per tenant:

Assignment Type	Description	Example
Shift	Fixed time block	Matatu shift (6am-2pm)
Day	Full day	Construction day (8 hours)
Hour	Hourly tracking	Event security
Task	Per unit completed	20 deliveries, 50 crates picked
Booking	Per customer transaction	10 bus tickets sold
Visit	Per location visit	15 home health visits

Capabilities:

- Start/end tracking (GPS, manual, or QR code)
- Units completed tracking (e.g., tons moved, parcels delivered)
- Multi-assignment per day (e.g., morning construction + evening security)
- Cross-tenant movement support (worker works for multiple employers)

4.3 Earnings Engine (Configurable)

Configurable earning models per tenant/role:

Model	Description	Example Industries
Hourly	Rate × hours worked	Construction, Event staff
Daily fixed	Flat rate per day	General labor, Security
Per unit	Rate × units completed	Logistics, Agriculture
Commission	% of revenue/value	Booking agents, Sales
Hybrid	Base + bonus/commission	Transport (fixed + commission)

Salary	Monthly fixed	Supervisors, Office staff
Attendance bonus	Extra for full attendance	Health promoters

Capabilities:

- Overtime rules (configurable multiplier, e.g., 1.5× after 8 hours)
- Daily aggregation across multiple assignments
- Multi-tenant aggregation (worker sees total earnings across employers)

4.4 Wallet & Payments

- Wallet ledger per worker
- Tenant float management (funds held for payouts)
- Automated payouts via JamboPay (daily, weekly, monthly — configurable)
- Transaction tracking (withdrawals, top-ups, transfers)
- Withdrawals to mobile money (M-Pesa, Airtel Money, etc.)

4.5 Payroll & Compliance (Configurable)

Statutory deductions (enabled per worker type):

- SHA (Social Health Authority)
- NSSF (National Social Security Fund)
- Housing Levy
- PAYE (where applicable)

Capabilities:

- Configurable deduction rules per tenant/worker role
- Payroll processing via Perpay
- Payslip generation (customizable per tenant)
- Statutory reporting templates (SHA, NSSF, KRA P-10, HELB)

4.6 Financial Services Layer

- Credit scoring (based on earnings consistency across assignments)
- Loan lifecycle management (application, approval, disbursement, repayment)
- Insurance enrollment & premium deduction
- Savings products (future)

4.7 Reporting & Analytics

- Earnings reports (per worker, work unit, tenant, industry)
- Attendance & assignment reports
- Financial summaries (float, payouts, fees)
- Compliance dashboards (per tenant)
- Industry-specific reports (e.g., transport vs construction)

5. TENANT CONFIGURATION FRAMEWORK

Each tenant (SACCO, contractor, logistics firm, etc.) configures AMY for their specific industry and workforce type.

Example Configuration: Transport (Matatu SACCO)

```
yaml
tenant_type: transport_sacco
assignment_unit: vehicle + route
roles: [driver, conductor, owner, booking_agent]
earning_models:
  driver: [fixed_shift, commission]
  conductor: [fixed_shift, commission]
  booking_agent: [commission]
payment_frequency: daily
statutory: [SHA, NSSF, Housing Levy]
facilitator_roles: [booking_agent, supervisor]
facilitator_payment: [commission_weekly, salary_monthly]
```

Example Configuration: Construction (Road Works Contractor)

```
yaml
tenant_type: construction
assignment_unit: site + section
roles: [general_labor, equipment_operator, foreman, site_supervisor]
earning_models:
  general_labor: [hourly, daily_fixed]
  equipment_operator: [hourly]
  foreman: [salary + attendance_bonus]
  site_supervisor: [salary_monthly]
payment_frequency: weekly (labor), monthly (supervisors)
statutory: [SHA, NSSF] # only for full-time roles
overtime: 1.5x_after_8hours
```

Example Configuration: Logistics (Delivery Platform)

```
yaml
tenant_type: logistics
assignment_unit: delivery_route
roles: [rider, dispatcher, sorter, warehouse_supervisor]
earning_models:
  rider: [per_delivery + bonus]
  dispatcher: [hourly]
  sorter: [hourly]
  warehouse_supervisor: [salary_monthly]
payment_frequency: daily (riders), bi-weekly (sorters), monthly (supervisors)
```

statutory: [SHA] # riders may opt out

Example Configuration: Community Health (NGO / Ministry of Health)

```
yaml
tenant_type: community_health
assignment_unit: clinic + village_coverage
roles: [promoter, lead_promoter, area_coordinator]
earning_models:
  promoter: [daily_stipend + per_visit]
  lead_promoter: [daily_stipend + attendance_bonus]
  area_coordinator: [salary_monthly]
payment_frequency: monthly
statutory: [SHA] # only for coordinators
tracking_method: gps_visit_confirmation
```

Example Configuration: Facilitator (Cross-Cutting)

```
yaml
worker_type: facilitator
description: "Non-primary workers who enable operations"
example_roles: [booking_agent, site_foreman, dispatcher, recruiter, supervisor]
earning_models: [commission, hourly, salary, hybrid]
payment_frequency_configurable: [daily, weekly, bi-weekly, monthly]
statutory_applicable: conditional_on_role
financial_profile_build: true # facilitators also need credit access
```

6. FEATURE SET

6.1 Worker App

Home

- Wallet balance
- Today's/today's shift earnings
- Active assignment (if any)

Earnings

- Daily / weekly / monthly view
- Per assignment breakdown (shift, day, task, visit)
- Per tenant breakdown (if working for multiple employers)

My Work

- Assignment history
- Work units (vehicles, sites, routes, villages)
- Hours / units completed

- Overtime tracked

Wallet

- Balance, withdraw, top-up
- Transaction history

Loans (Phase 3)

- Apply, track, repay

Insurance (Phase 3)

- Policies, claims

Deductions

- SHA, NSSF, Housing Levy (where applicable)

6.2 USSD (Feature Phone Access)

1. Check Balance
2. Withdraw
3. Today's Earnings
4. Last Payment
5. Loan Status (Phase 3)
6. Register

6.3 Tenant Admin Dashboard

Dashboard

- Payments overview
- Active workers (by role)
- Earnings summary (by work unit / site)

Workforce Management

- Add/edit/remove workers
- Role assignment (configurable per industry)
- KYC verification

Assignments (Core Feature)

- Assign workers to work units (vehicles, sites, routes, villages)
- Track shifts/hours/tasks
- Bulk assignment

Earnings (Core Feature)

- Per worker / work unit / tenant
- Approve/edit hours before payout (configurable)

Payroll

- Run, review, approve (frequency: daily/weekly/monthly)
- Select workers by role or work unit

Wallet Management

- Fund tenant float
- Payment tracking
- Payout reconciliation

Compliance

- Statutory reporting (per worker type)
- Deduction summaries

Configuration (Admin only)

- Earning models
 - Assignment types
 - Payment frequencies
 - Statutory rules
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7. USER STORIES (Industry-Specific)

Transport (Crew)

- As a driver, I want to track real-time earnings per shift so I know my daily income.
- As a conductor, I want to see commission breakdown so I understand my earnings.
- As a SACCO admin, I want to assign drivers to vehicles and routes so I can track operations.

Construction (Worker)

- As a general laborer, I want to clock in/out at the site so my hours are recorded.
- As a site foreman, I want to approve worker hours so payroll is accurate.
- As a contractor, I want to run weekly payroll for 200 workers so I avoid manual errors.

Logistics (Rider)

- As a delivery rider, I want to see earnings per delivery so I know my daily total.
- As a dispatcher, I want to assign riders to orders so deliveries are tracked.

Community Health (Promoter)

- As a CHP, I want to record my daily home visits so I earn my per-visit stipend.
- As a program manager, I want to see visit reports so I verify coverage.

Facilitator (Cross-Industry)

- As a booking agent, I want to earn commission on each booking so I am paid fairly.
- As a site foreman, I want a monthly salary + attendance bonus so I am motivated.
- As a supervisor, I want my office staff paid monthly while field staff are paid daily.

Lender

- As a lender, I want access to a worker's verified earnings history so I can assess credit risk (with consent).

Insurance Provider

- As an insurer, I want to deduct premiums automatically so coverage is consistent.
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8. FUNCTIONAL REQUIREMENTS

Module	Requirements
Identity	Unique worker ID (cross-tenant), KYC, role management, multi-tenant association
Assignment	Configurable types (shift, day, hour, task, visit, booking), multi-assignment per day, overlap handling
Earnings	Real-time calculation, configurable models (hourly, daily, per-unit, commission, hybrid, salary), aggregation across tenants
Wallet	Ledger, payouts, reconciliation, withdrawals, top-ups
Compliance	Configurable deduction automation, payroll via Perpay
Credit	Scoring (based on earnings consistency), loan tracking
Tenant Config	Industry-specific configuration (assignment types, earning models, payment frequencies)

9. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement
Performance	Real-time earnings updates; support for thousands of concurrent workers
Reliability	99.9% uptime; fault-tolerant payment processing (retry + dead-letter queues)
Security	Encryption at rest & in transit; RBAC; audit logs
Scalability	Horizontal scaling for multi-tenant, multi-industry growth
Offline Support	Capture assignments offline (mobile app); sync when online
USSD Constraints	<60s session timeouts; low-latency responses

10. INTEGRATIONS

Type	Integration	Purpose
Payments	JamboPay	Wallet management, payouts, float funding
Payroll	Perpay (Netcom)	Payroll processing, statutory calculations, payslips, P9 cards, SHIF/NSSF/KRA P-10/HELB reports
Identity	IPRS (Gov't)	KYC validation
Future	SHA, NSSF APIs	Direct statutory filing
Future	Banking APIs	Loan disbursement & recovery
Future	Insurance APIs	Premium remittance & claims
Future	Mobile Money (M-Pesa, Airtel)	Withdrawals and top-ups

11. RISKS & MITIGATION

Risk	Mitigation
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Tenant resistance (new industries)	Pilot with 2-3 tenants per industry; demonstrate ROI with transport success
Fraud (fake hours/assignments)	Assignment validation (GPS, QR, supervisor approval), audit logs
Payment failures	Retry logic + dead-letter queue; fallback manual reconciliation
Multi-tenant conflicts (worker works for multiple employers)	Central identity system; clear earnings segregation per tenant
Regulatory complexity (varies by industry)	Configurable compliance module; legal review per industry vertical

12. MVP SCOPE

Phase 1 (MVP — Configurable Core)

- Worker registry (KYC via IPRS)
- Configurable assignment tracking (minimum: shift, day, hour — others as config)
- Configurable earnings engine (hourly, daily fixed, per unit, commission — minimum)
- Wallet payouts via JamboPay
- USSD (basic: balance, withdrawal, last earnings, register)
- Worker App (basic: dashboard, earnings, wallet)
- Tenant Admin Dashboard (basic: workforce mgmt, assignments, earnings, wallet)
- **Tenant configuration layer** (admin sets assignment type, earning model, payment frequency)

Phase 2

- Payroll & compliance (Perpay integration)
- Statutory deductions (SHA, NSSF, Housing Levy — configurable per worker type)
- Reporting (per tenant, per industry)
- Facilitator role support (commission, salary, hybrid models)

Phase 3

- Credit scoring & loans
 - Insurance integration
 - Advanced assignment tracking (GPS geofencing, QR check-in/out)
 - Savings products
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13. FUTURE ROADMAP

- AI-based credit scoring (using earnings consistency)
 - Predictive earnings analytics
 - POS fare/integration (transport, retail)
 - Cross-border expansion (East Africa)
 - Industry-specific modules (agriculture, retail, hospitality)
 - Savings and investment products
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14. OPEN QUESTIONS

Question	Implication
How granular should tenant configuration be?	Trade-off between flexibility and system complexity
Can a worker belong to multiple tenants across industries?	E.g., drives matatu AM + works construction PM — yes, must support
How do we handle statutory deductions for mixed formal/informal roles?	Configurable per worker type; some may opt out
What is the pricing model? Per tenant, per worker, per transaction?	Affects go-to-market across industries
Do we need industry-specific compliance reports?	Construction vs transport vs health — likely yes
Data ownership: tenant vs platform vs worker?	Legal review required
Dispute resolution for earnings?	Workflow needed
Float shortage handling?	Notification and retry logic
Offline-first architecture depth?	Critical for construction sites and rural health workers

15. WHAT AMY IS (AND IS NOT)

AMY IS:

A configurable workforce financial operating system for Africa's informal and semi-formal economies.

- Every earning is traceable to an assignment
- Every assignment is verifiable
- Every worker can build a financial identity
- Every tenant configures for their industry

AMY IS NOT:

- Just a wallet
- Just payroll software
- Just a transport solution
- A one-size-fits-all system

Document Status: Ready for review

Next Steps: Validate with pilot tenants across transport, construction, and community health; finalize tenant configuration UI; begin Phase 1 sprint planning.

AMY: Because every worker deserves a financial identity.